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unthinkable categories and the fundamental laws of kinship

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> One eventually begins to wonder whether Nature herself does not abhor disjunctive groupings.
>
> —J.S. Bruner, J.J. Goodnow, and G.A. Austin, *A Study of Thinking*

In *L'Exercice de la parenté*, Françoise Héritier (1981) develops a general theory of kinship systems based on two "fundamental laws" *(lois fondamentales)*. These laws depend for their proof on a sociological explanation of two "unthinkable" *(impensables)*—that is, theoretically possible but empirically unrealized kinship terminologies implicit in the typologies of Lowie (1968[1929]) and Murdock (1949). Terray (1986) rejects this explanation on logical grounds but is unable to give any reason for the nonoccurrence of these two kinship terminologies. There is, however, a cognitive-linguistic explanation. According to Greenberg (1966, 1990[1980]), kinship terminologies, such as the unrealized one in Lowie's typology, do not occur, or are rare, because of a universal tendency to avoid disjunctive categories. With the notable exception of Nerlove and Romney (1967) and subsequently Kronenfeld (1974), Greenberg's theory has been overlooked by most anthropologists and is not mentioned in research works or discourses on kinship analysis (e.g., Barnard and Good 1984; Needham 1971).[1] This is unfortunate because, as Kroeber (1909) insisted and as Lowie (1968[1929]) conceded, there are certain aspects of kinship terminologies that can only be explained in cognitive-linguistic terms.

I have a double aim. The first is to evaluate Héritier's laws in the light of Greenberg's theory of universals in kinship classification. The second is to point out two important but unnoticed cognitive-linguistic features of Lowie's classification: his implicit lattice model of terminological change and his recognition that kinship terminologies may become simpler rather than more complex in the course of time. The lattice, or, more generally, semilattice model, first introduced by Epling, Kirk, and Boyd (1973) in the analysis of Polynesian sibling terms, and later applied by Hage and Harary (1996) to Oceanic sibling terms, shows how conjunctively defined kinship terminologies evolve through successive binary partition—in other words, through the successive addition or deletion of binary features. It captures multilinear as well as unilinear, and converging as well as diverging, paths in the evolution of all kinship terminologies. The model was in fact anticipated by Lowie in his classification of first ascending-generation terminologies.[2]

In L'Exercice de la parenté, *Héritier develops a general theory of kinship systems based on two "fundamental laws." These laws depend for their proof on a sociological explanation of two logically possible but empirically unrealized terminologies in the classifications of Lowie and Murdock. Following Greenberg's cognitive-linguistic theory of kinship classification, I interpret these unrealized terminologies in the first instance as special cases of a universal tendency to avoid disjunctive categories. In developing this interpretation I introduce an evolutionary perspective that has been absent from most discussions of kinship theory for a long time. The structural models for this analysis are, as always, graph theoretic.* [kinship theory, cognition, linguistics, graph theory]

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On the issue of the direction of evolutionary change, linguists (e.g., Blust 1984) recognize a tendency toward simplification under certain conditions, but anthropologists (e.g., Epling et al. 1973; Marshall 1984) often do not.[3]

Héritier's theory

Héritier's book constitutes, as Terray (1986) says, the most decisive progress in the analysis of kinship systems since the publication of Lévi-Strauss's *Les Structures élémentaires de la parenté* (1949). In intricately related arguments it explains the general laws underlying kinship terminologies, reveals the hidden correspondences between terminologies, descent modes, and marriage rules, and elucidates the functioning of semicomplex marriage systems and, by generalization, that of complex systems as well. Our concern is with the laws that underlie kinship terminologies.

The axiomatic basis of Héritier's theory consists of three biological facts: the duality of the sexes, the succession of generations, and differences within a generation due to birth order. Héritier's "first fundamental law of kinship" holds that "cross-solidarity is never stronger than parallel solidarity and a cross-relation between individuals or groups is never the implicit basis of equivalence or identity" (Héritier 1981:38, my translation). This law is expressed in the classification of relatives in the first ascending generation. According to Lowie there are four types of terminologies:[4]

1. Generational: F = FB = MB.
2. Lineal: F ≠ FB = MB.
3. Bifurcate merging: F = FB ≠ MB.
4. Bifurcate collateral: F ≠ FB ≠ MB.

There is, however, as Héritier points out, a fifth logically possible but empirically unrealized type not considered by Lowie:

5. $F \doteq MB \neq FB$.

According to Héritier this type does not occur because it is "unthinkable that the relation between two men linked through a woman, sister of one, wife of the other, could be closer than the relation between two brothers" (1981:42). She observes, however, that one could imagine a society in which this fifth possibility was realized. Under a rule of patrilineal descent and amitalocal residence, a rule that, as Murdock (1949) noted, is not found in any society, wives would take their Hs to reside with the brides' FZs. The head of the resulting amitaclan, the FZ, would be succeeded by her BD. The ideal marriage would be with FBD: this would permit a man to reside with his amitaclan and provide male assistance to the clan doyenne. Under this rule F could be classed with MB as "mother's kin," that is, as an affine or consanguine of M and distinguished from FB. Such an arrangement is "unthinkable" in that it "gives more weight, residential stability and superiority to the feminine principle than to the masculine principle. If inversely, a matrilineal avunculocal system is thinkable it is because it gives more weight, residential stability and superiority to the masculine principle" (1981:51). Héritier concludes from this thought experiment that the second fundamental law of kinship is male dominance and that this law is the "condition of the first."[5]

Terray (1986) objects that Héritier's thought experiment, although plausible, assumes a biconditional relation *(une correspondance bi-univoque)* between kinship terminology and social practice and therefore cannot be regarded as "proof" of her proposed laws of kinship:[6]

> The problem posed by such a simulation is that of the relation between terminology and social practice. This relation is neither simple nor direct; consequently, from a given terminology one cannot *deduce* any

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> definite practice. Inversely, from any practice whatever, one cannot infer *a* specific terminology; certainly the possibilities do not exist in unlimited quantity because there are laws of compatibility and incompatibility that reduce the number; but between terminology and practice there is never a biunique correspondence, and it can never be posited in advance that one is the simple reflection of the other. All things considered, Françoise Héritier concludes from her experiment in fictional anthropology that male dominance in the realm of practice prohibits the realization of certain combinations in the realm of terminologies: such an argument can be credited with a certain plausibility, but it certainly does not have the value of a proof. [Terray 1986:266, my translation]

Following the logic of Terray's objection, we note that there is not a biconditional relation between terminology and practice in the "inverse" of amitalocal residence, that is, between bifurcate merging terminology and avunculocal residence.

If social practice cannot account for the nonoccurrence of the fifth logically possible type of kinship terminology implicit in Lowie's classification, then what can? Barnard and Good (1984) call attention to this type but are unable to give any explanation for its nonoccurrence. In the symmetrical case of female terminologies in the first ascending generation, Murdock (1949) noted the rarity of the fifth type (M = FZ ≠ MZ) but otherwise offered no comment.[7] The answer was given many years ago by Greenberg (1966, 1990[1980]).

Greenberg's theory

In Greenberg's first paper on kinship classification (1966) he observed that the fifth logically possible terminology implicit in Lowie's typology does not occur because it groups together the two most divergent relatives, F and MB, without including FB. In his second paper, Greenberg (1990[1980]) analyzed the symmetrical case of female relatives in the parents' generation and specified that the fifth type (M = FZ ≠ MZ) does not occur because it contains a "disjunctive" category:

> The principle involved is the avoidance of logically disjunctive definitions. There is no way of demarcating by a single set of defining properties a term which embraces the mother and the [father's] sister without including in its reference the [mother's] sister. This is because mother's sister shares matrilineality with mother and collaterality with father's sister, but there is no common property of mother and father's sister, the two most different terms, except female, first ascending generation, and this includes mother's sister in its reference. [Greenberg 1990(1980):320]

In Bruner, Goodnow, and Austin's seminal work in cognitive psychology (1956) a *conjunctive category* is "one defined by the joint presence of the appropriate value of several attributes" (1956:41). A *disjunctive category* is one defined by the presence of one *or* another of these attributes.[8] In concept-attainment experiments in which a subject is required to learn categories defined by the attributes of figure (squares, circles, or crosses), color (red, green, or black), and number (one, two, or three figures), an example of a conjunctive category is one defined by three red circles. A disjunctive category is one that has "three red circles *or* any constituent thereof: three figures, red figures, circles, three red figures, red circles or three circles" (Bruner et al. 1956:41). Experiments show that disjunctive categories are much more difficult to learn than conjunctive ones. The reasons are that "one can predict only faultily or probabilistically from class membership to the properties of members [and that] two positive exemplars of a class may share no common characteristic" (Bruner et al. 1956:180). In addition, in learning a disjunctive category, one must rely on negative instances; in other words, one must determine what is not a member of a category.

Generalizing from these results in psychology to kinship terminologies in anthropology, Nerlove and Romney (1967) proposed that sibling terminologies be viewed "as an outcome of a 'natural' cognitive experiment." They predicted that disjunctive sibling terminologies would have "low survival value" and would therefore be extremely rare. Their hypothesis was dramatically confirmed: in a worldwide sample of 240 sibling terminologies only 4 were found to contain disjunctive categories.

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Greenberg, in proposing a cognitive-linguistic theory of kinship terminologies, considered the possibility of a sociological explanation for the unrealized type in Lowie's typology but rejected it on the grounds that the avoidance of disjunctive categories is a universal tendency in all classification systems, not exclusively those organizing kinship. He rejected an agnostic logical-positivist position for the same reason:

> To consider once more the Lowie-Kirchhoff typology,[9] the non-occurrence of the only logically disjunctive type was attributed to a cognitive principle. However, one might argue that this has rather to do with the nature of social institutions. There are no social arrangements of residence or descent which will align the mother and father's sister against the mother's sister. In general this argument could be used against the purely cognitive explanation of any disjunction. Alternatively, if inclined towards logical positivist type arguments, one could assert that the whole question is meaningless.
>
> There is, however, an argument, which is, I believe, decisive for the existence of a purely cognitive factor here, and that is that the same basic principles hold for other terminologies, such as color terms. A similar argument holds regarding marking hypotheses. For these also hold throughout other aspects of language including terminologies with other subject matters. [1990(1980):323–324]

It is convenient to represent the semantic structure of kinship terminologies using the graph theoretic model of a twin binary tree (Hage and Harary 1996). The model is more revealing than lists as in Greenberg (1966) and more immediate than spatial representations as in D'Andrade (1995). A conjunctive kinship category is one that can be represented by a single endnode of a twin binary tree while a disjunctive category is one that cannot. Greenberg defines the kinship terminologies in Lowie's typology by the dimensions (attributes) of (1) collaterality: lineal (ancestral) versus collateral (nonancestral)—l versus c; and (2) side: patrilineal (father's side) versus matrilineal (mother's side)—p versus m. Figure 1a shows twin binary tree models of the four types distinguished by Lowie plus the fifth logically possible but empirically unrealized type noted by Greenberg. The letters A, B, C represent native kinship terms (categories). The first four types have only conjunctive categories while the fifth type has a disjunctive category (or is disjunctive for short).

Héritier defines these terminologies slightly differently (Figure 1b). Instead of side she uses the dimension of parity: parallel (l l) relatives are parents and parents' same sex siblings while cross (x) relatives are parents' opposite sex siblings. As in Greenberg's analysis only the fifth "unthinkable" terminology is disjunctive.[10] Notice that in *this* case it does not make any difference which dimensions—parity or side—we use.

Grandparent terminologies provide a second kinship example of the tendency to avoid disjunctive categories. Greenberg lists the 15 logically possible grandparent terminologies as follows:

1. A. FaFa, FaMo, MoFa, MoMo
2. A. FaFa, FaMo B. MoFa, MoMo
3. A. FaFa, MoFa B. FaMo, MoMo
4. A. FaFa, MoMo B. FaMo, MoFa
5. A. FaFa B. FaMo, MoFa, MoMo
6. A. FaMo B. FaFa, MoFa, MoMo
7. A. MoFa B. FaFa, FaMo, MoMo
8. A. MoMo B. FaFa, FaMo, MoFa
9. A. FaFa, FaMo B. MoFa C. MoMo
10. A. FaFa, MoFa B. FaMo C. MoMo
11. A. FaFa, MoMo B. FaMo C. MoFa
12. A. FaMo, MoFa B. FaFa C. MoMo
13. A. FaMo, MoMo B. FaFa C. MoFa
14. A. MoFa, MoMo B. FaFa C. FaMo
15. A. FaFa B. FaMo C. MoFa D. MoMo

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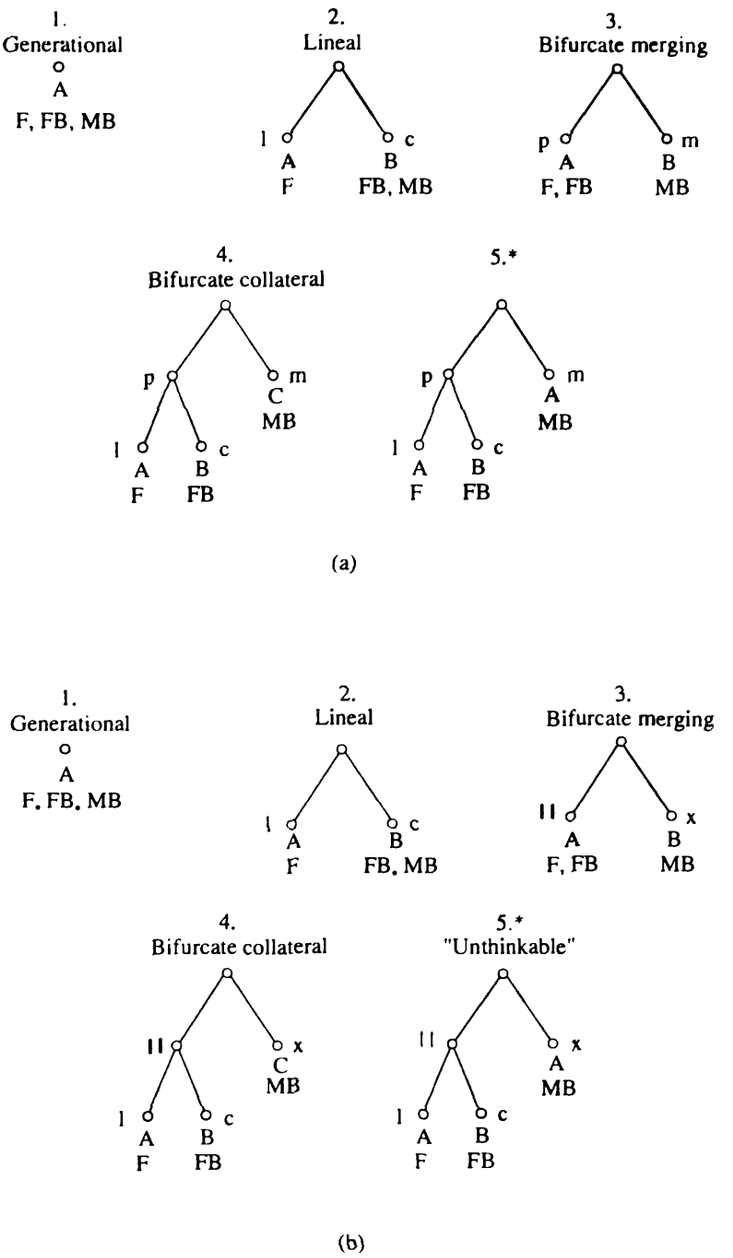


Figure 1. Conjunctive and disjunctive* first ascending generation kinship terminologies in (a) Greenberg and (b) Héritier.

In Greenberg's analysis of grandparent terminologies the parallel-cross distinction does not apply. Consequently Héritier's theory of identity and difference makes no predictions as to empirically unrealized types while Greenberg's theory does. The dimensions of contrast are sex of relative—male versus female (Δ versus O) and sex of connecting relative—male versus female (m versus f). Disjunctive terminologies are those equating FF and MM or MF and FM without including the other two kintypes. In Figure 2 the seven disjunctive terminologies are types 4*, 5*, 6*, 7*, 8*, 11*, 12*. Pooling his own sample of some 100 terminologies with Murdock's partially overlapping sample of 566 terminologies (Murdock 1970), Greenberg found that the disjunctive types either do not occur or are limited to one, two, or three societies.

Greenberg does not consider cousin terminologies. In contrast, Héritier does so in order to provide further evidence for the fundamental laws of kinship. Murdock (1949, 1970) distinguishes six types based on the classification of siblings (Sb = B, Z), parallel cousins (FBC, MZC), and cross-cousins (FZC, MBC) and on whether the patrilateral or matrilateral cross-cousin is raised a generation. Ignoring the raising of cross-cousins, which distinguishes Crow and Omaha types from the Iroquois type, and from each other, Héritier distinguishes four basic types of cousin terminologies:

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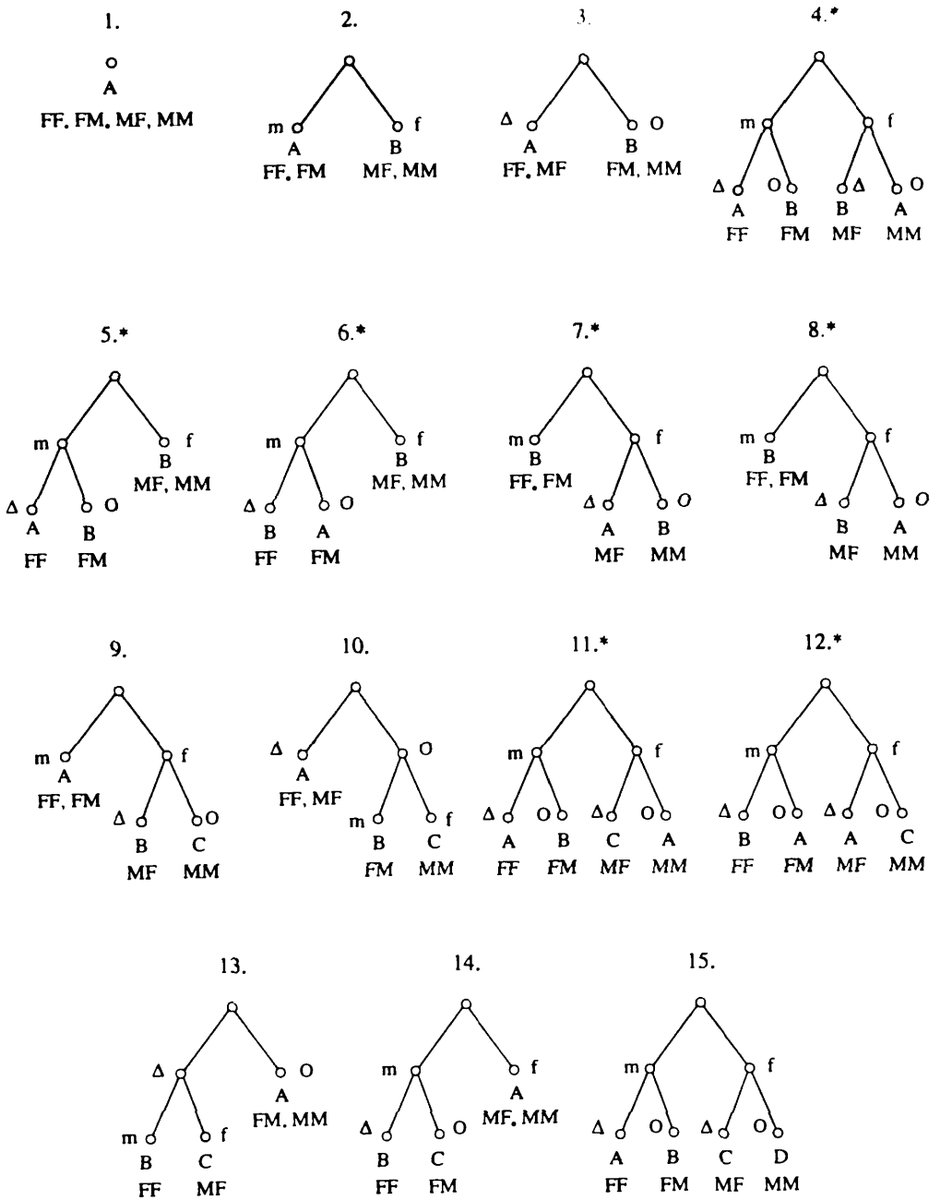


Figure 2. Conjunctive and disjunctive* grandparent terminologies.

1. Parallel cousins = cross-cousins = siblings (Hawaiian)
2. Parallel cousins ≠ cross-cousins ≠ siblings (Sudanese)
3. Parallel cousins = cross-cousins ≠ siblings (Eskimo)
4. Parallel cousins = siblings ≠ cross-cou sins (Iroquois, Crow, Omaha)

As with Lowie's typology, Héritier notes a fifth logically possible but empirically unrealized type that Murdock failed to consider:

5. Siblings = cross-cousins ≠ parallel cousins.

In Figure 3 these five types are depicted by twin binary trees. The dimensions of contrast are collaterality: first-versus-second degree (1 versus 2) and parity: parallel (siblings and children of parents' same-sex siblings) versus cross (children of parents' opposite sex siblings) (II versus x). The fifth type is disjunctive while the other four are conjunctive.

discussion

alternative analysis The grandparent terminologies could be analyzed from another point of view. In support of her theory that parity (parallel versus cross) is the basis of concepts of

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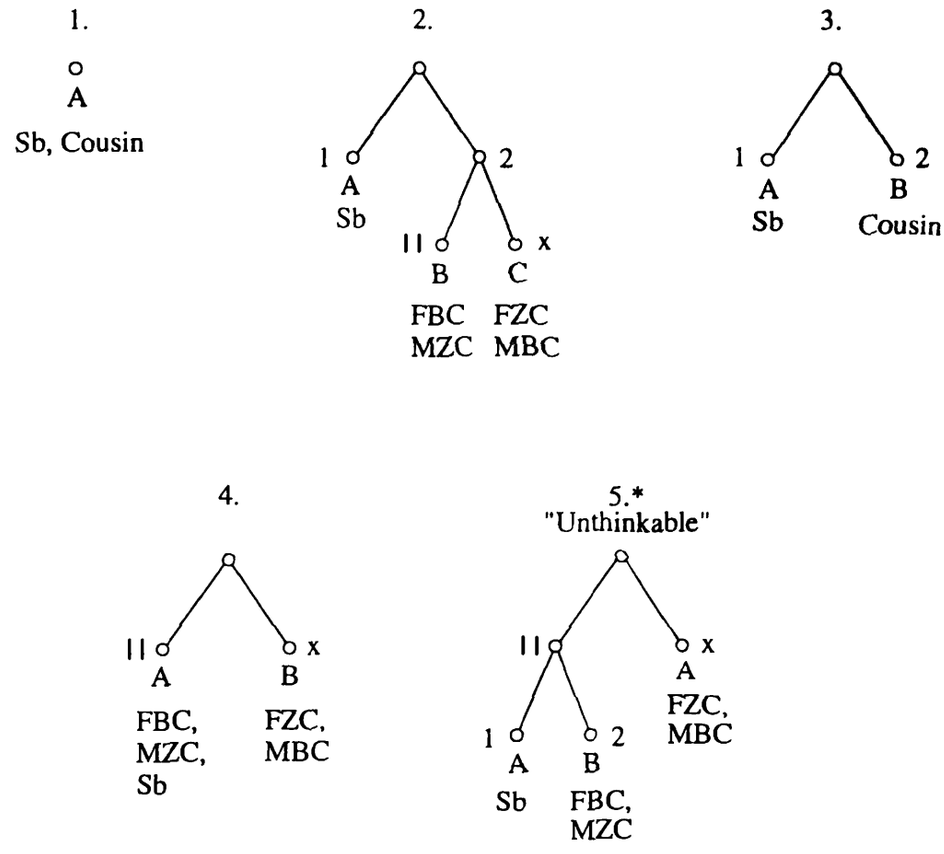


Figure 3. Conjunctive and disjunctive* cousin terminologies.

identity and difference in kinship systems, Héritier (1981, 1982) generalizes the cross-parallel distinction to relations of procreation.

> When procreation rather than siblinghood is privileged for the determination of the identical, commonality of gender still remains the primary criterion: societies more easily think of mother/daughter and/or father/son relations as privileged supports of identity than the cross-relation of father/daughter or mother/son. [1982:167]

Needham (1971) enumerates six logically possible modes of descent:

1. m → m (patrilineal)
2. f → f (matrilineal)
3. (m → m) + (f → f) (bilineal)
4. (m → f) + (f → m) (alternating)
5. (m → m) II (f → f) parallel
6. m/f → m/f (cognatic).

Although both the fourth and fifth types are rare and uncertain, Héritier maintains that the latter, based on same-sex links, is "easier to envisage" (1981:43) than the former, based on alternating-sex links.[11] Following this line of thought, one could apply the parity distinction to commonality of gender in grandparent terminologies. Scheffler (1987), in fact, does this in a critique of Greenberg's theory of marking rules. Scheffler observes that grandparents could be classified as parallel or cross depending on whether the relative in question is the same or opposite sex as the linking relative. Thus a terminology that classifies together FF and MM as against FM and MF, type 4 in Figure 2, is disjunctive according to Greenberg but conjunctive according to Scheffler (as shown in Figure 4). As Greenberg observes (1987b), Scheffler only adduces a single terminology of this type. On the other hand, type 3, which is conjunctive according to Greenberg, would be disjunctive according to Scheffler. But type 3 is the most common grandparent terminology of all. Similarly, type 11, which would be conjunctive

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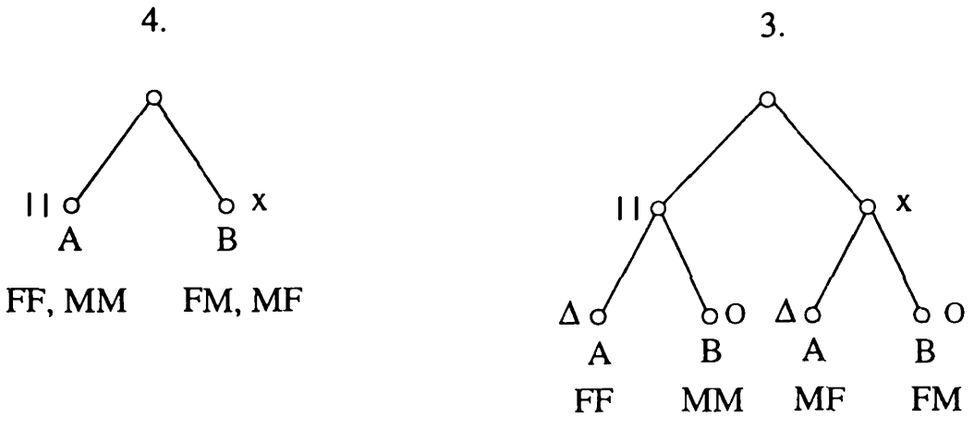


Figure 4. Reclassification of grandparent terminologies using the dimension cross/parallel.

according to Scheffler but disjunctive according to Greenberg, does not occur at all. There is thus no support for Héritier's laws in relations of procreation.

marking rules All four of the logically possible, conjunctive first ascending-generation kinship terminologies in Lowie's typology are empirically realized, but this is not true in general. In Greenberg's theory there are three determinants of realizable kinship terminologies: the avoidance of disjunctive categories, salience, and marking. Salience refers to a hierarchical relationship between different dimensions of contrast. For example, sex of relative is more salient than sex of connecting relative "in that systems which neutralize the former but not the latter are infrequent whereas those which neutralize the latter but not the former are extremely common" (Greenberg 1990[1980]:321).

Marking refers to a hierarchical relationship between values of the same dimension. The unmarked term of an opposition is more general than the marked term. Marking applies to all languages and to every level of a language—phonological, grammatical, and lexical. In English and other European languages, for example, the present tense is unmarked as against the past, and the term *man* is unmarked as against the term *woman*.[12] In kinship, according to Greenberg, the marked term has three characteristics:[13]

1. Overt expression. For example, in English kinship terminology the term "grandparent" is marked while "parent" is unmarked.
2. Fewer distinctions. In English the sex distinction is neutralized in the marked category "cousin" as opposed to the categories "brother" and "sister."
3. Defectivation: the marked term lacks an intersecting category. In English there are terms for "brother-in-law" and "sister-in-law" but not for "cousin-in-law."

Summarizing a wealth of cross-cultural evidence, Greenberg (1990[1980]:318) proposes that "lineal is unmarked as against collateral, consanguineal is unmarked as against affinal, male is unmarked as against female in regard to sex of referent, older is unmarked in relation to younger. . . . In general, the closer a generation is to ego, the more unmarked it is." We say "proposes" because, as Greenberg (1987b:369) emphasized in reply to a criticism by Scheffler (1987), the validity of these criteria "has not been taken for granted, since every application is in effect also a test." One such test was made by Nerlove and Romney (1967), who used marking hypotheses derived from Greenberg (together with constraints of cognitive economy) to reduce the logically possible 146 conjunctive sibling terminologies to the 12 that occur with significant frequency.

This is not to say that Greenberg's theory rules out sociological determinants of kinship terminologies. For example, "Crow and Omaha systems generally, perhaps always, require matrilineal and patrilineal descent respectively" (1990[1980]:324). But even here linguistic

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factors come into play. In regard to Lounsbury's rejection (1964) of the unity of the lineage theory of Crow and Omaha terminologies (Radcliffe-Brown 1952), Greenberg observes:

> It has been noted, for example by Lounsbury, that in most instances not a single kin term is in fact confined to relatives in a particular unilineal group. It may also be noted that the speaker's unilineal group always contains more distinct terms than the lineage of a father in a Crow system or mother in an Omaha system. Both of these are understandable in terms of marking theory. The lack of correlation between clan membership and terminology is typically with more remote relatives where there is in all systems obliteration of distinctions. Further, given the importance of unilineal grouping, it will form a basis for distinguishing terminology but again the principle asserts itself that the more remote grouping is less differentiated. [1990(1980):324]

evolutionary semilattices In his paper on kinship terminology Lowie did more than merely distinguish among four different types. He related each type to worldwide geographical areas and he also considered possible evolutionary changes in kinship terminologies, conceiving of such changes in binary terms and as proceeding from complex to simple rather than conversely (as often happens with linguistic structures). Thus Lowie observed:

1. "The [bifurcate merging] system, which Morgan and Rivers genetically connect with the [generation] system (though in a reverse sense) is not one whit further from the [bifurcate collateral system]" (1968[1929]:42).
2. In Asian systems, "it is evident that emphasis on status and seniority would cause a drift from the bifurcate merging toward the generation pattern" (1968[1929]:49).
3. "Morgan was not warranted in assuming that the Polynesian term translated [as] 'father' implied procreation: the facts simply reveal a common status term for all kindred of one generation. Simplicity here, as in many other linguistic phenomena, is a late development, not a badge of antiquity. Three independent investigators, Sternberg in Siberia, Rivers in Oceania, Lowie in North America, have been able to show recent transformations of bifurcate merging nomenclatures in the direction of generation systems" (1968[1929]:56).
4. "The designations here employed are awkward but serve to bring out connections usually disregarded. Specifically, Morgan and Rivers stressed the genetic relationship of Generation and Bifurcate Merging terminologies: Morgan traced the development of the latter from the former; while Rivers reversed the process. The recognition of Bifurcate Collateral terminologies opens a new prospect—the derivation of the Bifurcate Merging from the Bifurcate Collateral type. Logically, the affiliation is not one iota smaller than between the two systems compared by Morgan and Rivers. . . . When, for some reason—say the joint presence of the levirate and the sororate—partial merging develops, the Bifurcate Merging type would come into being" (1928:266–267).

The evolutionary changes described by Lowie have the structure of a "lattice" (Hage and Harary 1996). In a digraph (directed graph) *D* (Harary et al. 1965), an *upper bound w* of two nodes *u* and *v* in *D* is a node that can reach both *u* and *v* by directed paths. Further, *w* is a *least upper bound (LUB)* for *u* and *v* if it is not only an upper bound, but, for any other upper bound *w'*, every path from *w'* to either *u* or *v* must contain *w*. The directional dual of the LUB *(u, v)* is their *greatest lower bound (GLB),* defined similarly.

A *lattice* is a partially ordered set in which every two nodes have both an LUB and a GLB. A *semilattice* is a partially ordered set in which every two nodes have an LUB. It follows that every lattice is a semilattice but not vice versa. In this sense a semilattice is a more general mathematical structure than a lattice.

It is customary to represent a lattice (or a semilattice) by its "Hasse diagram." In a digraph *D* node *u covers* node *w* if there is no node *v (uvw).* In the *Hasse diagram* of a lattice only the arcs of its covering digraph are drawn. It is also a convention to omit the directional arrows but I will retain them for clarity.[14]

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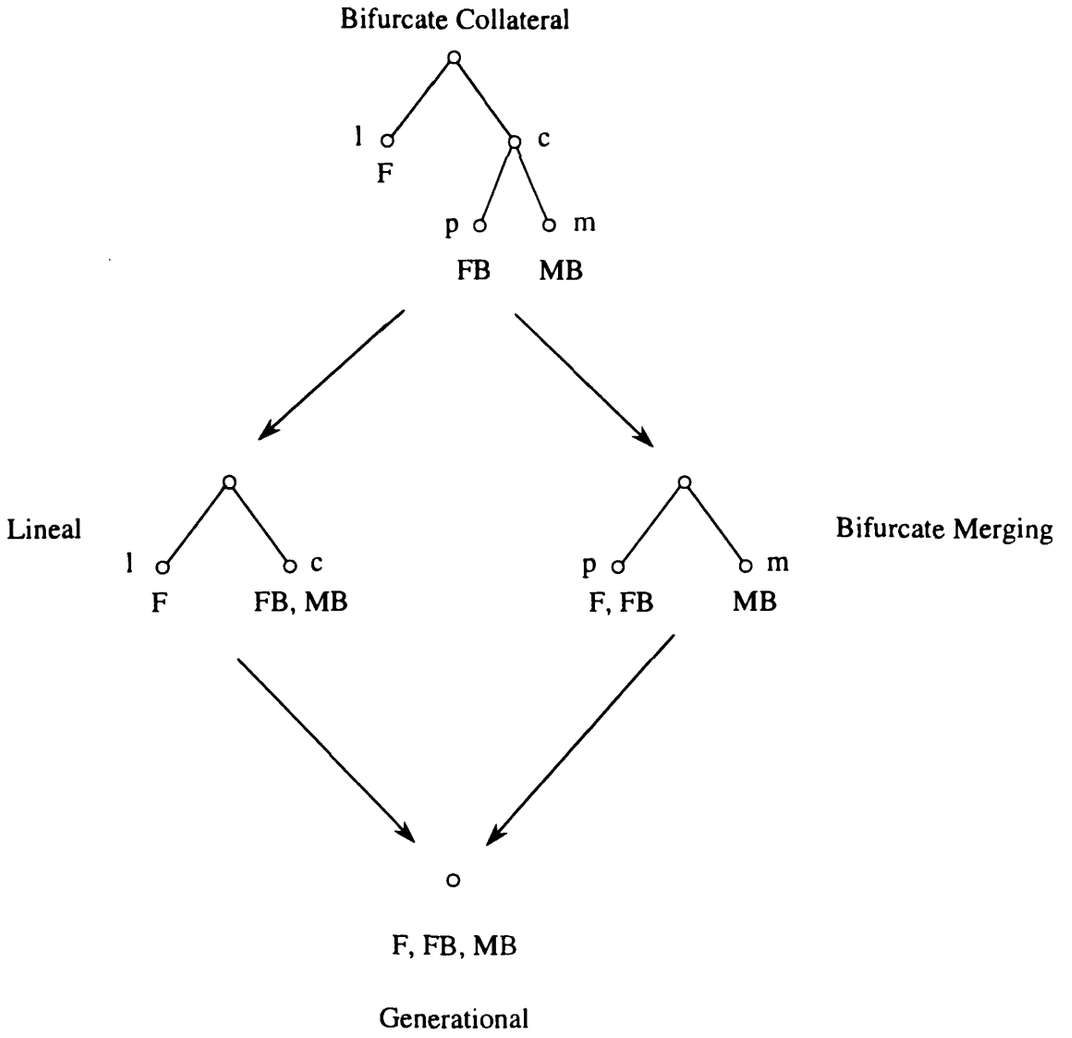


Figure 5. A lattice model of the evolution of first ascending generation kinship terminologies.

Evolutionarily interpreted, a semilattice is a general model that can take the form of a directed path (a "unilinear" structure), an oriented branching tree (a "multilinear" structure), a digraph that diverges and converges, or one that diverges and partially converges.

Lowie's hypotheses concerning the evolution of kinship terminologies are clearly summarized and depicted in the lattice of Figure 5. The nodes represent kinship terminologies modeled as twin binary trees and the arcs represent evolutionary changes through successive deletions of binary dimensions of contrast (this time using Greenberg's dimensions). The pattern is one of divergence and convergence.

An example of a lattice (i.e., the Hasse diagram) having the form of a directed path comes from Polynesia. It illustrates (1) the need for linguistic data in studying the evolution of kinship terminologies, (2) Lowie's point about the tendency toward simplification in terminological change, and (3) the historical effect of marking rules. In the Polynesian languages there are four types of sibling terminologies. Those consisting of more than a single term are conjunctively defined on one or more of the dimensions of sex (male versus female—m versus f), seniority (elder versus younger—e versus y), and parity (where parallel and cross refer to same- and opposite-sex siblings respectively, l l versus x). The terminologies, which have one, two, three, or four terms, are:

1. Sibling.
2. Parallel versus cross-sibling.
3. Parallel sibling; male versus female cross-sibling.
4. Elder versus younger parallel sibling; male versus female cross-sibling.

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In a purely typological (nonlinguistic) analysis Epling, Kirk, and Boyd (1973), following Sahlins's theory (1958) of the development of social stratification in Polynesia, assumed that sibling terminologies evolved from simple to complex, in the order just listed. In a subsequent typological study of sibling terminologies in Island Oceania, Marshall (1984) argued for the same sequence on the grounds of frequency and geographical distribution. The actual sequence was discovered by Clark (1975). Using the traditional methods of historical linguistics Clark reconstructed Proto-Polynesian sibling terms and concluded that the evolutionary sequence was just the reverse of the one proposed by Epling, Kirk, and Boyd (1973) and by Marshall (1984). This sequence, for terminologies having cognate terms, is shown in Figure 6. As in Lowie's model of first ascending-generation terminology, change occurs through the progressive deletion of binary features. (We note that the lattice in Figure 6 is a subdigraph of a much larger diverging and almost completely converging semilattice that includes *all* the reported sibling terminologies in the Oceanic language family [Hage and Harary 1996]).

Clark speculated that an explanation for the "pattern of semantic survival" in Figure 6 in which one term replaces another "might be sought in terms of semantic markedness" (1975:86). If replacement is treated as the diachronic equivalent of neutralization—that is, the leveling of oppositions—then the unmarked terms are *male, younger,* and *parallel.* The first replacement is predicted by Greenberg's theory of marking, although the second is not.[15] The second

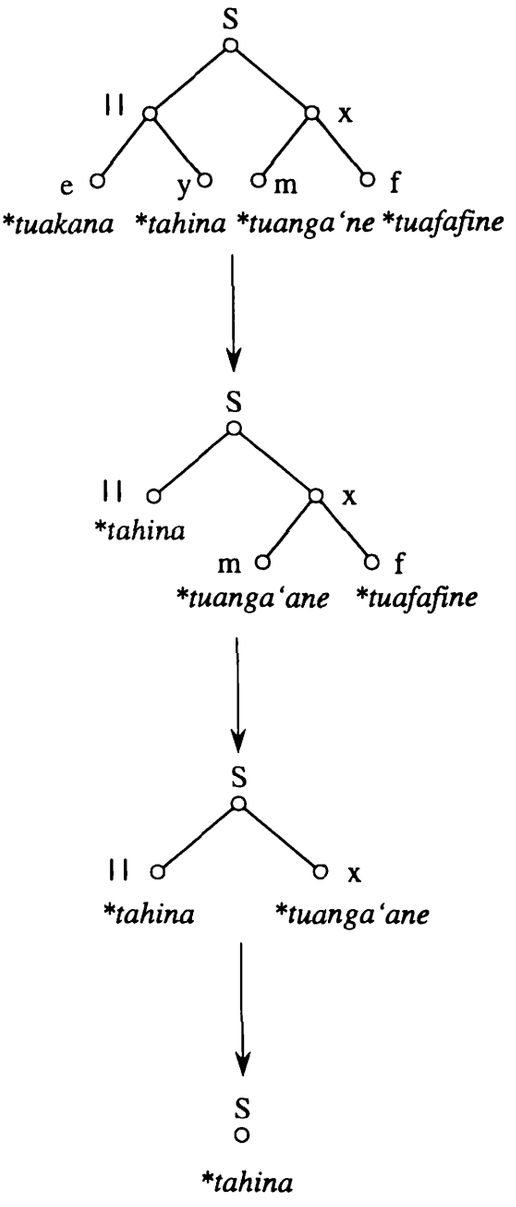


Figure 6. A lattice model of the evolution of Polynesian sibling terminology based on Clark (1975) from Hage and Harary (1996).

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replacement is, however, consistent with Scheffler's observation (1987) that elder sibling is marked in many Malayo-Polynesian languages. The third replacement supports Scheffler's hypothesis that "where the contrast 'same' vs 'opposite' (i.e., parallel vs cross) is treated as an opposition there is a strong tendency to treat 'same' as unmarked" (1987:219). This suggests that parity should be added to Greenberg's list of marking universals in kinship terminologies.

conclusion

Following Greenberg's theory we conclude that the unthinkable kinship terminologies in Lowie's and Murdock's typologies should be viewed *in the first instance* as special cases of a cognitive-linguistic tendency to avoid disjunctive categories. This tendency is found in all kinship terminologies, including those not based on the cross-parallel distinction. It is so widespread in classification systems generally that one could speak of the mind's "abhorrence of disjunctiveness" (Bruner et al. 1956). In cognitive anthropology the avoidance of disjunctive categories was the original motivation and the basic assumption of the formal analysis of kinship terminologies. As Lounsbury said in his pioneering paper on the subject,

> A componential definition represents a Boolean class product, and is thus a "unitary" or "conjunctive" definition. It is assumed that the meaning of any term belonging to a properly defined paradigm—one whose semantic field is itself unitary—will be susceptible to such a definition. This is perhaps a stronger item of faith than we have a right to hold at this moment; but it furnishes the motivation for the analysis of kinship systems at least. . . . Were we to compromise on this point and admit disjunctive definitions (class sums, alternative criteria for membership) as on a par with conjunctive definitions (class products, uniform criteria for membership), there would be no motivation for analysis in the first place, for definitions of kin classes by the summing of discrete members . . . are disjunctive definitions par excellence. [1968(1964):127]

In kinship analysis it may not be possible to avoid some disjunctive categories, but that is always the aim. That is why Goodenough (1980), for example, prefers his own analysis of Kariera terminology to that of Romney and Epling (1958) and Reid (1967). In everyday life, disjunctive categories are avoided when possible, sometimes by being made to appear conjunctive. Kronenfeld (n.d.) gives the example of treating siblings as lineal ("direct") relatives "even though they are not directly on ego's ancestor/descendant line by conceiving of them as on the ancestor/descendant line of ego's nuclear family and not noticing that ego's ancestors and descendants, respectively, pertain to different nuclear families—one of orientation and procreation."

In scientific research, as Bruner, Goodnow, and Austin observe, the discovery of a disjunctive explanation "seems to act as a challenge to future research rather than as an occasion to celebrate a 'finding' " (1956:162). In cross-cultural studies, for example, such a challenge is evident in attempts to explain bifurcate merging kinship terminology. In Murdock's reading (1947), Rivers (1914) explained bifurcate merging terminology by the presence of moieties in a society. Lowie (1915, 1917, 1919) "supplemented" this explanation by showing its relation to exogamy. Finally, Kroeber (1917) attributed bifurcate merging terminology to the presence of unilineal descent groups. In effect, Kroeber gave a more general, conjunctive explanation that "subsumes the essential factors dealt with by Rivers and Lowie" (Murdock 1947:62).

The subject of this paper goes back to a classic argument in anthropology as to whether kinship terms are primarily determined by linguistic and psychological factors, as Kroeber (1909) held, or by sociological conditions, as Rivers (1914) maintained. In attempting to strike a balance between these positions, Lowie concluded that "there will always be residual phenomena resisting interpretation on any but linguistic lines. This means that they are in a sense unique facts that can be understood after they have been observed but that could not be deduced from general principles" (1968[1929]:59). It is clear from Greenberg's theory that cognitive-linguistic

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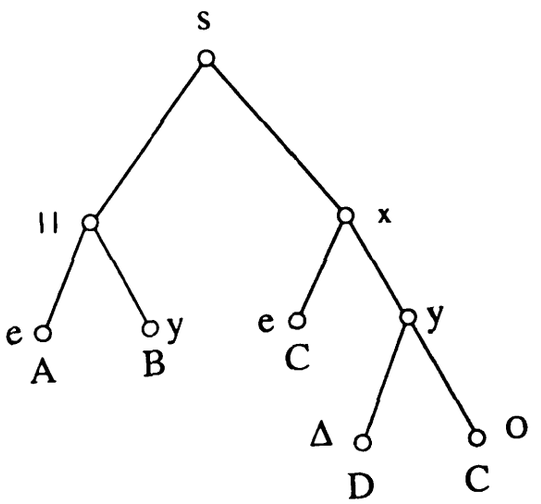


Figure 7. An unthinkable Polynesian sibliing terminology.

facts are primary, deducible phenomena. They demarcate the boundaries of empirically realizable kinship terminologies.

In Polynesian sibling terminologies, for example, the seniority distinction reflects an emphasis on primogeniture (Pawley 1980), while the parity distinction is consistent with the presence (or former presence) of unilineal descent groups (Murdock 1968); but these dimensions are not free to combine in any arbitrary way. A hypothetical terminology like the one in Figure 7 would express these important social distinctions; but it is disjunctive and therefore unthinkable—that is to say, it does not occur in any of the 15 Polynesian sibling terminologies based on these distinctions.

notes

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This article is dedicated to the distinguished graph theorist, Frank Harary, on the occasion of his 75th birthday.

1. Greenberg's theories of linguistic classification (1963, 1987a), on the other hand, are well-known in physical anthropology and archaeology (e.g., Cavalli-Sforza 1991; Cavalli-Sforza et al. 1988; Cavalli-Sforza et al. 1994; Greenberg et al. 1986; Renfrew 1992; Ruhlen 1987).

2. Implicit uses of semilattices as evolutionary models of kinship structures include Lévi-Strauss's "classification of the principal types of kinship systems evolved from dual organization" (1949) and Needham's "scheme of transformations" (1984) of symmetrical and asymmetrical prescriptive marriage systems in eastern Indonesia.

3. In general, kinship terminologies may evolve in both directions. Color terms and biotaxonomies, on the other hand, generally become more complex (Berlin 1972; Berlin and Kay 1969). In language, changes at different levels may interact so that simplification at one level may result in complexity at another level. See, for example, Goodenough's analysis (1992) of phonological and morphological change in Chuukese (Trukese).

4. I use the conventional kinship notation: F = father, M = mother, B = brother, Z = sister, C = child, S = son, D = daughter, with FB = father's brother, FBC = father's brother's child, and so forth.

5. Matrilocal residence would appear to be an exception to this rule. But Héritier notes, following Lévi-Strauss (1949), that the number of matrilineal matrilocal societies is extremely small and that many matrilineal societies are in fact avunculocal.

6. The *biconditional* statement, written $p \leftrightarrow q$, can be read "if *p* then *q*, and if *q* then *p*." It is also written *p* iff *q*, and read "*p* if and only if *q*."

7. The symmetric parent-aunt cases are included in Lowie (1928). Murdock found five cases of this type, so it is almost but not completely unthinkable.

8. The *disjunctive* statement $p \vee q$ may be read *"either p or q is true." Inclusive* disjunction means *"p or q or both"* while *exclusive* disjunction means *"p or q but not both."*

9. Lowie (1928) and Kirchhoff (1931) independently invented this typology.

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10. One of the reviewers has called our attention to the fact that a logically possible fifth type of Crow-Omaha terminology is empirically unrealized for reasons parallel to the disjunctivity argument presented in this paper. See Hammel (1965).

11. Barnard (1975) reports a certain but rare case of alternating descent.

12. Marking is pervasive and extends even to mathematics and logic. "Thus negative is always taken as the marked member of the positive-negative opposition; -5 is always negative, but 5 by itself is either the absolute value of 5—that is, 5 abstracted from its sign value—or +5 as the opposite of the marked negative category. So in logic *p* is used ambiguously either as the proposition *p* abstracted from its truth value as either true or false or, on the other hand, for the assertion of the truth of *p*" (Greenberg 1966:25–26).

13. Greenberg (1987b) gives ten criteria of markedness applicable to different aspects of language.

14. A detailed explanation and illustration of these concepts is given in Hage and Harary 1996.

15. Greenberg qualifies his first prediction:

> it may well be that neither male nor female can be described as the unmarked category on a universal basis. In a number of instances the male term has zero expression where the corresponding female term has an additional morpheme, but the data on neutralizations give conflicting evidence. [1966:79]

In Seneca, female is unmarked (Lounsbury 1968[1964]). On contextual variation in marking see Waugh 1982.

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